KneeGait-Edge: Standalone Hardware Build & Biomechanical Guide

Self-powered edge-AI wearable knee monitoring system | ZINNOVATIO 4.0 Hackathon | Revised build guide

Sampling	Radio	Power	Sensor site	Output
50 Hz, on-device complementary filter	BLE 5.0 (Wi-Fi off)	3.7 V LiPo, USB-C rechargeable	Anteromedial tibia, single IMU	Peak swing angle per stride + stiffness alert

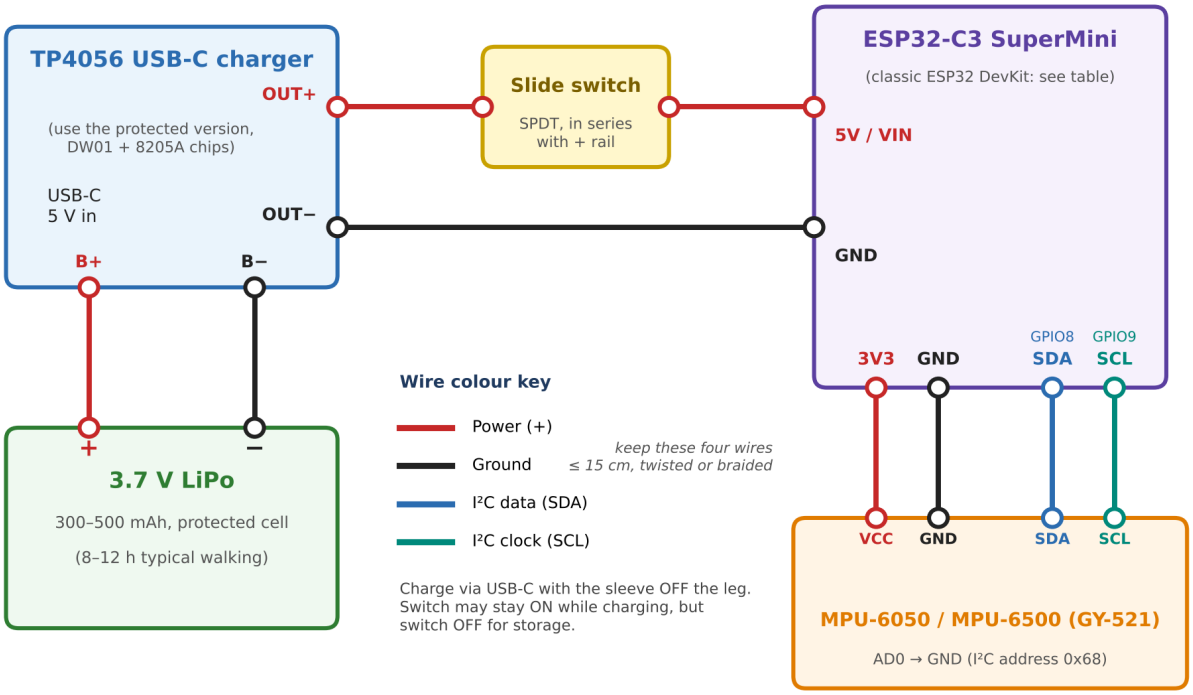
1. Bill of materials (DIY)

Component	Specification	Purpose and buying notes	Est. cost
Microcontroller	ESP32-C3 SuperMini (preferred) or ESP32 DevKit	BLE 5.0, 50 Hz loop, Wi-Fi switched off to save power. The C3 SuperMini is far smaller; the DevKit is easier to breadboard.	₹250 – ₹350
IMU	MPU-6050 or MPU-6500 breakout (GY-521 / GY-6500)	6-DOF accelerometer + gyro. Set ±2 g and ±500 °/s ranges. Breakout includes I²C pull-ups and a 3.3 V regulator.	₹120 – ₹180
Battery	3.7 V LiPo, 300 – 500 mAh, **with protection circuit**	Flat pouch cell. Runtime 8 – 12 h is an estimate for BLE streaming (~25 – 40 mA average); measure it (see §6).	₹150 – ₹220
Charger	TP4056 USB-C module, **protected version (DW01 + 8205A)**	CC/CV charging from any 5 V source. The protected board adds over-discharge and short-circuit cut-off, so connect the load to OUT+/OUT−.	₹35 – ₹50
Power switch	Mini SPDT slide switch	Hard disconnect in the + rail so the cell does not drain in storage.	₹10
Wearable	Elastic knee sleeve with silicone patella ring + 25 mm elastic strap	Ring self-centres on the patella. Many sleeves end above the sensor zone, so add a strap or choose a longer sleeve.	₹150 – ₹250
Casing and wire	3D-printed box or pillbox, 30 AWG silicone wire, heat-shrink, Kapton tape	Sweat-proof, rounded corners, insulated battery compartment. Flexible wire resists walking strain.	₹50
Total			**₹765 – ₹1,110**

Prices are approximate Indian-market estimates; check current listings before ordering.

2. Electrical wiring and circuit schematic

The module is self-contained. The LiPo cell connects to the charger's B+/B− pads. The charger's OUT+ feeds the ESP32 5 V/VIN pin through the slide switch, and OUT− is the common ground. The cell recharges through the USB-C port without opening the casing.



From	To	Signal / function
LiPo (+)	TP4056 B+	Cell connection and charging input
LiPo (−)	TP4056 B−	Cell ground
TP4056 OUT+	Switch pin 1; switch pin 2 → ESP32 5V/VIN	Switched, protected supply rail (≈3.0 – 4.2 V from the cell)
TP4056 OUT−	ESP32 GND	Common ground
ESP32 3V3	MPU-6050 VCC	Regulated 3.3 V to the IMU
ESP32 GND	MPU-6050 GND	Sensor ground
ESP32 SDA pin*	MPU-6050 SDA	I²C data, 400 kHz fast mode (100 kHz also works)
ESP32 SCL pin*	MPU-6050 SCL	I²C clock
MPU-6050 AD0	GND	Fixes I²C address at 0x68

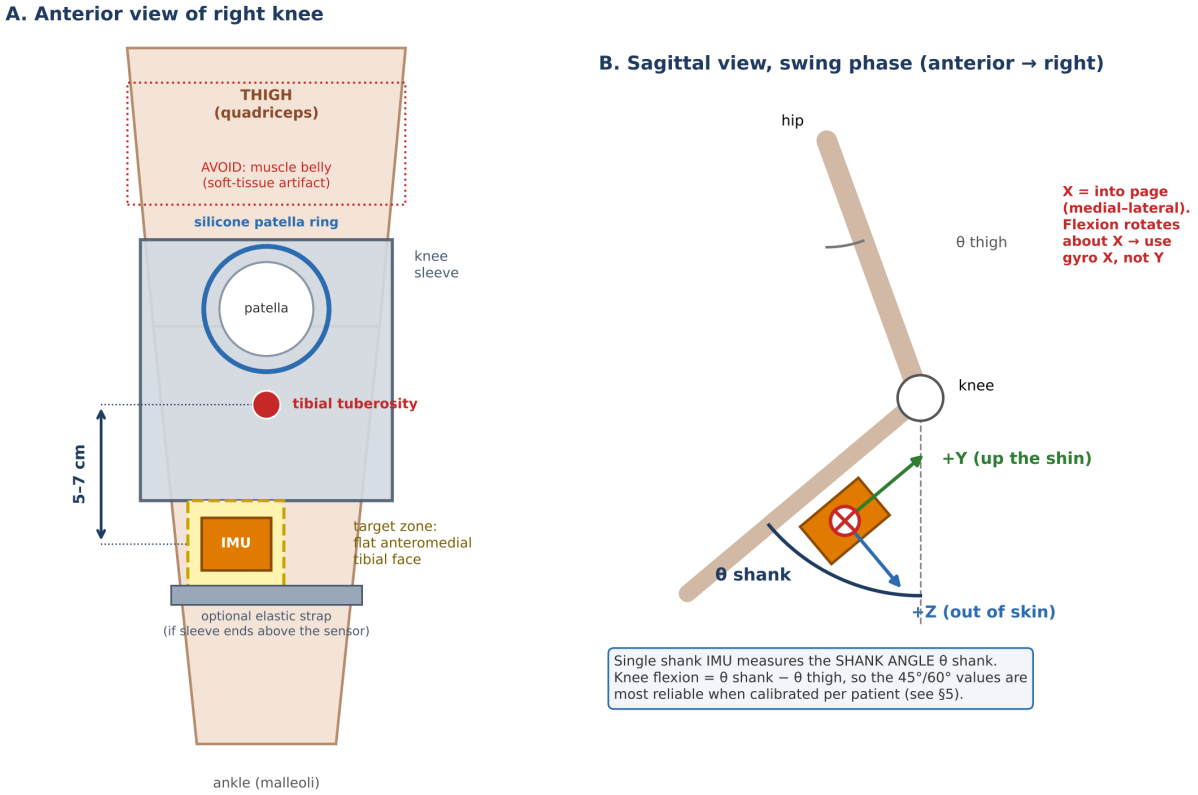
Board	SDA	SCL	Note
ESP32-C3 SuperMini	GPIO8	GPIO9	Arduino `Wire` defaults. GPIO21/22 do **not** exist as I²C pins on this board.
ESP32 DevKit (classic)	GPIO21	GPIO22	Arduino `Wire` defaults.

*Any free GPIO works if you call `Wire.begin(SDA, SCL)` explicitly.

Power notes. The cell swings from 4.2 V (full) to about 3.0 V, and the board's onboard regulator needs roughly 3.5 V or more to hold 3.3 V, so the device may reset a little before the cell is empty. Feed the 5V/VIN pin only; never connect the cell directly to the 3V3 pin.

3. Clinical anatomical mounting and soft-tissue rejection

Accurate gait capture depends on rigid skeletal coupling. The figure shows the landmarks and sleeve alignment (A) and the sensor axes with the quantity actually measured (B).



Anatomical feature	Protocol and justification
1. Patella ring anchor (self-centring)	The sleeve's silicone ring seats over the kneecap when the sleeve is pulled up. This removes the need for anatomical training and keeps placement repeatable between sessions.
2. Tibial target zone	Feel for the tibial tuberosity (the bony bump below the kneecap where the patellar tendon inserts). Centre the sensor **5 – 7 cm (2 – 3 in) below it** on the flat anteromedial (inner-front) face of the tibia. Mark the position with skin-safe tape on the first fitting.
3. Soft-tissue artifact (STA)	A thigh sensor sits on the quadriceps, which moves relative to the femur at heel strike, adding error to joint-angle estimates. The anteromedial tibia has very little soft tissue over it and no muscle belly, so the sensor stays close to the bone. (Quantitative STA figures: see the references note in §8.)
4. Axis alignment	Seat the IMU flush in the pocket with **+Y along the shin pointing up** and +Z pointing out of the skin. Flexion/extension is rotation about the **medial–lateral X axis**, so use **gyro X** as the rate input and **atan2(a_z, a_y)** as the accelerometer angle. Gyro Y would measure twist about the shin, not flexion.
5. Fit	Snug but not tight. The case must not rock; if it does, add a foam shim or a strap. Skin should be dry and intact, and the case must not press on the tibial crest.

4. Assembly steps

1. **Bench-test first.** Wire the ESP32 and IMU on a breadboard, power over USB, and confirm the I²C scanner finds 0x68 and that raw readings change when you tilt the board.
2. **Check the charger.** Confirm polarity with a multimeter before connecting the cell. LiPo polarity reversal can destroy the board or start a fire.
3. **Wire the power path** as in §2: cell → B+/B−, OUT+ → switch → 5V/VIN, OUT− → GND. Insulate every joint with heat-shrink.
4. **Connect the IMU** with four short (≤15 cm) wires, twisted together. Add a small strain-relief loop at each solder joint.

5. **Check current** with a USB power meter or multimeter. Expect roughly 20 – 40 mA while advertising and streaming. Much higher means Wi-Fi is still on or a short exists.
6. **Enclose.** Place the battery flat at the bottom of the box, the board above it with a thin foam layer between, and leave the USB-C port and switch accessible. Keep the IMU rigid in the case, with its axes marked on the lid.
7. **Mount in the sleeve** following §3, then run the calibration in §5.

5. Operating protocol and signal processing

- **Power-up and auto-zero.** Switch on and stand still for 2 s. The device averages the gyro output to get the zero-rate bias.
- **Neutral-angle calibration.** Stand upright for another 2 s with the knee straight to record the angle offset; subtract it from all later angles.
- **50 Hz sampling.** A complementary filter ($\alpha = 0.96$, $dt = 0.02$ s, time constant $\approx \alpha \cdot dt/(1-\alpha) = 0.48$ s) fuses gyro and accelerometer angles to give the sagittal shank angle with limited drift.
- **Telemetry.** Batch about 5 samples per BLE notification (≈10 notifications/s) to cut radio on-time. Suggested sample: timestamp (uint32), angle ×100 (int16), gyro ×10 (int16).
- **Dashboard.** The Vite web companion connects over the Web Bluetooth API (Chrome or Edge, HTTPS or localhost) and plots the angle trace and stride metrics.

```
const float ALPHA = 0.96f, DT = 0.02f;           // 50 Hz loop
float angle = 0;                                  // degrees, shank from vertical

void step() {                                     // call every 20 ms
  float rate  = gx - gyroBiasX;                   // deg/s about medial-lateral X
  float accel = atan2f(az, ay) * 57.2958f;        // inclination from gravity
  angle = ALPHA * (angle + rate * DT) + (1 - ALPHA) * accel - angleOffset;
}
// If the angle falls when the foot swings back, flip the sign of 'rate' and 'accel'.
```

Stride segmentation and alert

- Detect each swing peak of the angle trace with a minimum peak prominence (e.g. 15°) and a refractory period of about 0.4 s so one stride gives one peak.
- **Metric:** peak swing angle per stride, averaged over a rolling window of 5 – 10 strides.
- **Alert threshold:** literature values for peak knee flexion in swing are about 60° for healthy gait, and the default alert is 45°. That is **75 %** of the healthy reference. A single shank sensor reports the shank angle, not the knee angle, so set the alert at **75 % of the wearer's own baseline** measured in a calibration walk, and use 45° only as a starting default.
- **Naming.** Persistently reduced swing flexion is usually called *stiff-knee gait*. *Antalgic gait* means a pain-avoiding pattern with a shortened stance phase, so label the alert "reduced swing flexion / stiff-knee pattern" unless you also detect stance asymmetry.

6. Validation checklist

Test	How	Pass criterion (suggested)
Static angle	Hold the shank at 0°, 30°, 60°, 90° on a protractor or phone inclinometer	Error ≤ 2° at each angle
Drift	Stand still for 5 min	Angle drift ≤ 1° over 5 min
Dynamic	Walk 20 strides on a flat corridor while filming (side view) and compare peak swing angle per stride	Mean error ≤ 5° against video-based angle
Repeatability	Remove and refit the sleeve 5 times	Peak angle varies ≤ 3° between fittings
Battery life	Stream continuously from a full charge until shutdown	Record hours; report the measured value, not the estimate
BLE range and loss	Walk 5 m from the phone/laptop with the sensor on the shank	Packet loss < 1 %

7. Safety and troubleshooting

- **Battery safety.** Use only a protected LiPo cell in a rigid, insulated case. Never charge it while worn, never puncture or bend it, and stop using it if it swells or gets hot.
- **Skin.** Keep the case edges rounded, fabric between case and skin, and wear sessions short for the first tests. Remove it if there is redness or pressure marks.
- **Scope.** This is a research prototype. It is not a certified medical device and its alerts should not be used for diagnosis or treatment decisions.

Symptom	Likely cause	Fix
I²C scan finds nothing	SDA/SCL swapped, wrong GPIO for the board, or loose joint	Check the pin table in §2; reflow joints; call `Wire.begin(SDA, SCL)`
Angle drifts steadily	Gyro bias not zeroed, or the sensor moved during auto-zero	Re-run the 2 s still calibration; keep α near 0.96
Angle jumps or is noisy during swing	Loose case, or the accelerometer picking up linear acceleration	Add foam or a strap; raise α to 0.98 if jitter remains
Angle moves the wrong way	IMU mounted rotated or flipped	Re-check axes (§3) and flip the sign in firmware
Battery lasts only 2 – 4 h	Wi-Fi left on, or one BLE notification per sample	Disable Wi-Fi, batch samples, lengthen connection interval
Device resets when you start walking	Cell voltage dips or a loose power joint	Add a 100 µF capacitor across 5V/GND; check solder and switch contacts

8. References and claims to verify

The original guide cited **Song & Baxter 2025** (STA noise >60 %, tissue under 2 mm) and **Bytyqi et al. 2014** (alert validated at 99.85 % accuracy). I could not confirm these here, so check each against the source paper before presenting them. The 99.85 % figure in particular describes that study's own setup; do not present it as the accuracy of this device. Report your own results from the §6 tests instead, and add the citations with full author lists, journal and DOI once verified.